

Conditional adjacency anonymity in social graphs under active attacks

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Abstract Social network data is typically made available in a graph format, where users and their relations are represented by vertices and edges, respectively. In doing so, social graphs need to be anonymised to resist various privacy attacks. Among these, the so-called active attacks, where an adversary has the ability to enrol sybil accounts in the social network, have proven difficult to counteract. In this article we provide an anonymisation technique that successfully thwarts active attacks while causing low structural perturbation. We achieve this goal by introducing $(k, \Gamma_{G,\ell})$ -adjacency anonymity: a privacy property based on (k, ℓ) -anonymity that alleviates the computational burden suffered by anonymisation algorithms based on (k, ℓ) -anonymity and relaxes some of its assumptions on the adversary capabilities. We show that the proposed method is efficient and establish tight bounds on the number of modifications that it performs on the original graph. Experimental results on a collection of randomly generated graphs show that, when compared to methods based on (k, ℓ) -anonymity, the new method continues to provide protection from equally capable active attackers while introducing a much smaller number of changes in the graph structure.

Keywords Social network privacy, graph perturbation methods

1 Introduction

Online social networks (OSNs) have become the most successful digital application of our time. Over two billion persons¹ regularly use some OSN to interact with friends, share information, etc. As a result of this, massive amounts of information about personal relationships, consumption patterns, personal preferences, and

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¹ Source: statista.com.

more, are generated every day. An important part of this information is encoded in the form of social graphs. In a social graph, a vertex corresponds to a person (a user of the OSN), whereas edges represent relations between individuals. Personal information, such as name and email address, is usually associated to vertices as attributes. Edges can also have associated attributes.

This massive amount of information is enormously valuable. OSNs themselves analyse this data in order to determine the advertisements they show to their users, filter out information that they consider not to be relevant etc. As data holders, the OSN can effectively access the totality of the available data. However, third parties, such as social scientists, market researchers and public institutions are also interested in accessing and analysing part of this information for conducting population studies, assessing the effect of communication campaigns, surveying public opinion, and many other purposes. In order to enable these studies, it is necessary that OSN administrators release samples of their social graphs. However, despite the usefulness of the studies that can be conducted on the released data, the sensitive nature of the information encoded in social graphs raises serious privacy concerns.

In releasing a social graph for analysis, an indispensable first step for preserving user privacy is to remove all personally identifiable information from its vertices and edges. However, as shown in [14], even a graph with no identifying attributes can leak sensitive information, since some structural properties (e.g. the degree and neighbours of a vertex) can be unique to certain users. A *re-identification attack* seeks to leverage some knowledge about a set of users, the victims, to re-identify them after the graph is released. For example, an adversary who knows the number of friends of a victim can later re-identify her in the released graph if such value happens to be unique. Once a set of users is re-identified, the attacker can learn sensitive information, such as the existence of relations between two users or the affiliation of some of them to a community.

According to the means by which adversaries obtain the knowledge used to re-identify the victims, they are classified as *passive* or *active* [1]. On the one hand, a passive adversary relies on existing information, obtainable from publicly available sources, such as other OSNs, but does not attempt to purposely alter the structure of the network. On the other hand, an active adversary creates a set of accounts in the network and links them to other users, in such a manner that each victim is induced to uniquely satisfy some structural property allowing the adversary to re-identify all the victims after publication of the graph.

A common approach in counteracting re-identification attacks is to transform the original graph into another (similar) graph that satisfies a given privacy property. In this approach, the original graph is first stripped of any vertex and edge attributes, obtaining a pseudonymised graph, which is isomorphic to the original graph, but user information is replaced by uninformative, randomly generated labels. Then, a series of vertex/edge additions and removals are performed until the privacy property is satisfied. To that end, a number of privacy properties, based on the well-known notion of k -anonymity [18], have been proposed. The first such property proposed for counteracting passive attacks was k -degree anonymity [9], which protects social graphs from adversaries that know the degrees of the victim vertices. The authors of [9] provided an efficient algorithm for obtaining k -degree anonymous graphs with a reasonable utility preservation, and later authors have proposed utility-oriented improvements on this type of methods [10, 2, 5, 11, 17].

Additionally, new privacy properties, that account for stronger passive attacks, have been proposed. For example, k -neighbourhood anonymity [23] allows to protect a social graph from adversaries who know the subgraphs induced by the neighbours of their victims. Properties that account for increasing amounts of structural knowledge have been proposed, up to k -automorphism anonymity [24], which subsumes the rest. In all cases, every strengthening of the attacker model has come at the cost of obtaining anonymisation methods that alter the graph structure to a larger extent [24].

Regarding active attacks, a single privacy property, named (k, ℓ) -anonymity [19], has been proposed for quantifying the resistance of a graph to this type of attack. But this property has similar issues to k -automorphism [24]; both are remarkably hard to evaluate and enforce. Indeed, current transformation approaches aiming at (k, ℓ) -anonymity [12] considerably affect the structure of the original graph. As we will discuss in the following sections, some aspects of this measure can be better tailored to serve as the basis for anonymisation methods.

The goal of this article is to explore further into the problem of counteracting active attacks on social graphs via edge set transformations with low structural perturbation. Taking (k, ℓ) -anonymity as our starting point, we focus on two main issues. First, it is computationally expensive to evaluate (k, ℓ) -anonymity and to use it as the basis for anonymisation methods. Second, (k, ℓ) -anonymity seems to overestimate the capabilities of active attackers. It assumes that the adversary is able to determine the exact distances to arbitrary nodes in the network, which is infeasible in practice. This is evidenced by current active attacks [1, 15, 16], which are based on the neighbour relation between the victims and the attacker nodes.

Summary of contributions:

- We first identify an important limitation of (k, ℓ) -anonymity that has been overlooked so far, and makes it computationally expensive to use (k, ℓ) -anonymity as the privacy goal for perturbation-based anonymisation techniques, as we illustrate in Section 2 and validate in Section 5. As a fix for this limitation, we propose a new privacy property called $(k, \Gamma_{G, \ell})$ -anonymity (Section 2).
- Secondly, we introduce the notion of (k, ℓ) -adjacency anonymity, a well-motivated relaxation of the adversary capabilities assumed by (k, ℓ) -anonymity (Section 3). Adjacency anonymity considers that the adversary only has prior knowledge about the vicinity of the sybil nodes, rather than about the whole graph as assumed in [19, 12]. Conceptually, (k, ℓ) -adjacency anonymity relates to (k, ℓ) -anonymity as k -neighbourhood relates k -automorphism. By combining the two notions introduced in Sections 2 and 3, we obtain the new privacy property $(k, \Gamma_{G, \ell})$ -adjacency anonymity.
- We show that there exist efficient graph transformations towards $(k, \Gamma_{G, 1})$ -adjacency anonymity for large values of k (Section 4), while previous anonymisation techniques [12] can only guarantee (k, ℓ) -anonymity for some $k > 1$ or some $\ell > 1$.
- Finally, we perform experiments on random graphs (Section 5) to show that $(k, \Gamma_{G, 1})$ -adjacency anonymous graphs can be obtained with low impact on the structure of the original graph. Our experiments also confirm that $(k, \Gamma_{G, 1})$ -adjacency anonymous graphs are as resistant to some known active attacks as (k, ℓ) -anonymous graphs obtained by existing methods.

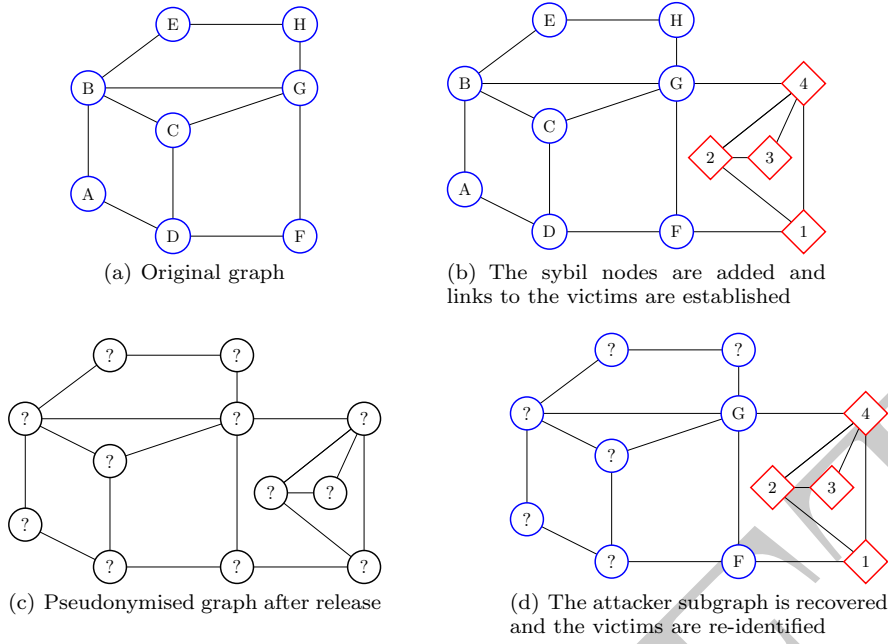


Fig. 1 The four stages of an active attack.

2 Revisiting (k, ℓ) -anonymity

In this section we review the notion of (k, ℓ) -anonymity and observe a fundamental limitation of this privacy property that makes it difficult to be used in perturbation-based privacy-preserving techniques. We define a revised privacy notion, called $(k, \Gamma_{G, \ell})$ -anonymity, to alleviate this limitation.

2.1 Active attacks

An active adversary relies on the ability to register several (fake) accounts to the network. Such accounts are called *sybil accounts* or *sybil nodes*, depending on whether one refers to the social network account or the social graph vertex. Prior to the publication of the social graph, the active attacker adds a set of sybil accounts to the network (e.g. nodes 1, 2, 3 and 4 in Figure 1(b)). The sybil accounts establish links between themselves and also with the victims (e.g. users F and G in Figure 1(b)). At this stage, the attacker only knows about the edges incident to the sybil nodes. Thus, she is unaware of the edge linking F and G .

After publication of the pseudonymised social network graph, the attacker first searches for the subgraph induced by the sybil nodes. Victims connected to the sybil subgraph can be re-identified by using the neighbour relation between sybil nodes and victims. For example, the non-sybil nodes connected to 1 and 4 in Figure 1(d), must be F and G , respectively. This allows the adversary to acquire

knowledge that was supposed to remain private, such as the existence of a link between F and G .

From a practical point of view, active attacks require the ability to create new accounts in the social network, which is trivial, and remain unnoticed by sybil detection techniques such as those described in [22]. From a theoretical point of view, an active attack relies on creating a uniquely identifiable attacker subgraph, which requires the subgraph induced by the sybil nodes to be isomorphic to no other subgraph in the network. Backstrom et al. [1] described the properties of the attacker subgraph that guarantee uniqueness (with high probability) and showed that, in most networks, $\log n$ sybil nodes, where n is the number of vertices, are sufficient for creating an attacker subgraph that compromises any vertex and goes unnoticed to sybil detection methods.

Effectively determining whether a social graph is vulnerable to an active attack is a necessary step towards developing a mitigation strategy against it. For example, the complete graph G satisfies that for every proper subgraph S there exists another subgraph S' that is isomorphic to S . This property obviously makes an active attack unfeasible in a complete graph. However, determining the actual resistance of an arbitrary graph to active attacks is not trivial. A first step in this direction was given in [19], where Trujillo-Rasua and Yero introduced the privacy measure (k, ℓ) -anonymity, which we describe in detail next.

2.2 (k, ℓ) -anonymity

Let $G = (V, E)$ be an arbitrary pseudonymised graph. Throughout this paper we will treat V as a set of randomly generated labels bearing no relation with the original identities of the network users. We will also assume that the elements of V can be traversed in some fixed (but arbitrary) total order $\preceq$ and will denote an ordered subset of vertices $(S, \preceq)$, $S \subseteq V$, simply as $S = \{v_1, v_2, \dots, v_t\}$, where $v_i \in S$ for $i \in \{1, \dots, t\}$ and $v_1 \preceq v_2 \preceq \dots \preceq v_t$. Whether the notation refers to sets or ordered sets will be clear from the context.

Given an ordered set of sybil nodes $S = \{s_1, \dots, s_t\}$ in G , Trujillo-Rasua and Yero [19] proposed to model the adversary knowledge about a vertex $u \in V$ using her distances to the sybil nodes. To that end, they use the vector

$$r_G(u \mid S) = (d_G(u, s_1), \dots, d_G(u, s_t)),$$

where the distance $d_G(u, v)$ is computed as the number of edges in a shortest path joining u and v . This vector is referred to as the *metric representation* of u with respect to S .

An active attacker is assumed to be able to re-identify those vertices having unique metric representations with respect to the set of sybil nodes under her control. In consequence, the data owner must ensure that every vertex in the published graph is undistinguishable from at least a minimum number of other vertices. This property is captured in [19] by the notion of *k -antiresolving set*, which is enunciated as follows.

Definition 1 (k -antiresolving set) Let $G = (V, E)$ be a non-trivial graph. A set $S \subseteq V$ is a *k -antiresolving set* of G if k is the largest positive integer such that,

for every $v \in V(G) \setminus S$, there exist vertices $w_1, w_2, \dots, w_{k-1} \in V(G) \setminus S$ such that $v, w_1, w_2, \dots, w_{k-1}$ are pairwise different and

$$r_G(v \mid S) = r_G(w_1 \mid S) = \dots = r_G(w_{k-1} \mid S).$$

Note that, if the set S of sybil nodes is forced to be k -antiresolving, then for every victim node v , there are at least $k-1$ other nodes that cannot be distinguished from v by only inspecting their metric representations with respect to S . The notion of (k, ℓ) -anonymity was introduced in [19] to quantify the privacy of a social graph in the presence of active attackers by considering that every set of vertices up to a given cardinality may potentially be a set of sybil nodes. The formal definition of (k, ℓ) -anonymity relies on the parameter k -metric antidimension. The k -metric antidimension of a graph G is the minimum cardinality of any k -antiresolving set of G .

Definition 2 ((k, ℓ) -anonymity) A graph G is said to satisfy (k, ℓ) -anonymity if k is the smallest positive integer such that the k -metric antidimension of G is smaller than or equal to ℓ .

From a privacy perspective, if a graph satisfies (k, ℓ) -anonymity, an attacker with the capacity to enrol, and successfully retrieve, up to ℓ sybil nodes in the graph would still be incapable of distinguishing any vertex from at least $k-1$ other vertices. Certainly, a graph satisfying (k, ℓ) -anonymity for $k > 1$ effectively resists active attacks when performed by at most ℓ sybil nodes. However, even the simplest of the privacy goals, namely transforming a $(1, 1)$ -anonymous graph into a (k, ℓ) -anonymous graph G' for some $k > 1$ or some $\ell > 1$, has not been accomplished without significantly disrupting the graph structure [12]. Our observation is that (k, ℓ) -anonymity, although suitable to quantify resistance against active attacks, cannot be applied straightforwardly to privacy-preserving transformation of social graphs.

2.3 A limitation of (k, ℓ) -anonymity

Privacy measures based on k -anonymity all rely on the same basic principle: every user should belong to one or more *anonymity sets* of size at least k , where an anonymity set is a collection of indistinguishable objects with respect to a given adversary. In (k, ℓ) -anonymity, given a graph $G = (V, E)$ and a subset $S \subseteq V$ with $|S| \leq \ell$, the set of anonymity sets defined by S is the set of equivalence classes in $V \setminus S$ with respect to the equivalence relation

$$R_{G,S} = \{(u, v) \in (V \setminus S) \times (V \setminus S) : r_G(u \mid S) = r_G(v \mid S)\}.$$

We use $[u]_S$ to denote the equivalence class, i.e. the anonymity set, of a vertex u with respect to $R_{G,S}$. Next we formalise the relation between anonymity sets and (k, ℓ) -anonymity, making it explicit that a necessary condition for a graph to satisfy (k, ℓ) -anonymity is that the cardinality of all anonymity sets of each user must be at least k .

Observation 1 A graph $G = (V, E)$ satisfies (k, ℓ) -anonymity if and only if k is the largest positive integer such that for every subset of vertices $S \subseteq V$ with $|S| \leq \ell$ and every vertex $u \in V \setminus S$ it holds that $|[u]_S| \geq k$.

Most privacy properties define a single anonymity set for each user. For example, k -degree anonymity considers an adversary that knows the number of links of the victims. Thus, the anonymity set of a vertex u in k -degree anonymity is the (unique) set containing all vertices with the same degree as u . In (k, ℓ) -anonymity, however, every user may belong to a number of anonymity sets that exponentially grows with ℓ .

The drawback of considering many anonymity sets for each user in a privacy property is twofold. First, the computational complexity of determining whether the property is satisfied increases with the number of anonymity sets. This problem has already been investigated by Chatterjee et al. [3], who showed that the k -metric antidimension problem used in (k, ℓ) -anonymity is NP-Hard. Second, a graph transformation approach should take into account that increasing the size of the anonymity set of a user u with respect to a subset of vertices S may decrease the size of the anonymity set of the same user with respect to a different subset S' . We illustrate this second drawback through the example in Figure 2.

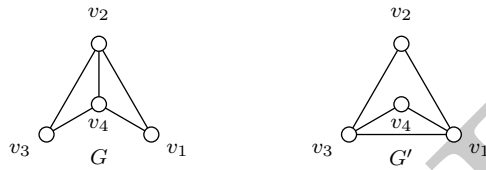


Fig. 2 An example of a counterintuitively unsuccessful anonymisation according to (k, ℓ) -anonymity.

The graph G in Figure 2 is $(1, 1)$ -anonymous, as the anonymity set of v_1 with respect to $\{v_3\}$ is $\{v_1\}$. Likewise, the anonymity set of v_3 with respect to $\{v_1\}$ is $\{v_3\}$. Thus, any transformation of G aiming to achieve $(2, 1)$ -anonymity should necessarily increase the cardinality of those two anonymity sets. That goal is achieved in the graph G' shown in the figure. However, in this graph the cardinality of the anonymity set of v_2 with respect to $\{v_4\}$, and vice-versa, has dropped from 3 in G to 1 in G' . Therefore, according to (k, ℓ) -anonymity, G' is an unsuccessful anonymisation of G . We argue that in the example above G' should actually be considered as a successful anonymisation of G , as it transforms all 1-antiresolving sets of G , i.e. the ones that originally posed a privacy threat, into 3-antiresolving sets. We elaborate on this idea in what follows.

2.4 $(k, \Gamma_{G, \ell})$ -anonymity

We introduce the rationale that the probability of success of an adversary does not increase after perturbation. In fact, the first stage of an active attack (see Figure 1(b) for reference) can only be considered successful if the attacker managed to make her set of sybil nodes a 1-antiresolving set of the original graph. Otherwise, the attacker will have more than one victim associated to the same fingerprint. In this circumstance, metric representations alone do not allow her to unambiguously assign the identity of any of those victims to any vertex of the graph after publication. For example, in Figure 2, even though $\{v_2\}$ is a 1-antiresolving set of G' ,

the re-identification probability of an adversary leveraging this set of sybil nodes in G is at most $1/3$ after the publication of G' , because $\{v_2\}$ is a 3-antiresolving set of G .

According to this rationale, we will consider that those vertex subsets that are k' -antiresolving, with $k' \geq k$, in the original graph, are **not** the result of a successful first stage of an active attack. In consequence, we will focus on protecting the published graph from the remaining vertex subsets. With this idea in mind, the goal of the remainder of this section is to establish a theoretical framework that allows to disregard subsets of vertices with low re-identification power in the original graph. Because (k, ℓ) -anonymity does not allow to model such a conditional protection, we first introduce a parametrisable privacy notion and then derive from it an instance that will allow us to model the desired conditional privacy protection.

Definition 3 ((k, Γ)-anonymity) Let G be a non-trivial graph and let $\Gamma \subseteq \mathcal{P}(V(G))$ be a family of subsets of $V(G)$. The graph G satisfies (k, Γ) -anonymity if every $S \in \Gamma$ is a k' -antiresolving set of G , with $k' \geq k$.

Clearly, (k, ℓ) -anonymity is an instance of (k, Γ) -anonymity, where $\Gamma = \{S : S \subseteq V(G), |S| \leq \ell\}$. More importantly, (k, Γ) -anonymity allows us to determine whether a graph G' , obtained by a series of perturbations on an initial graph G , reduces the threat potential of those attacker subgraphs in G with high probability of re-identification. We do so by instantiating (k, Γ) -anonymity with a concrete interpretation of Γ , denoted $\Gamma_{G, \ell}$, which contains all subgraphs in G of cardinality up to ℓ that are k' -antiresolving sets in G with $k' < k$. That is to say, given the privacy parameter k , the set $\Gamma_{G, \ell}$ accounts for all potential attacker subgraphs whose probability of re-identification is not guaranteed to be at most $\frac{1}{k}$. This instantiation is formally defined as follows.

Definition 4 (($k, \Gamma_{G, \ell}$)-anonymity) Let $G = (V, E)$ be a non-trivial graph and let $G' = (V, E')$ be the result of a series of perturbations on G . The graph G' is said to satisfy $(k, \Gamma_{G, \ell})$ -anonymity if it satisfies (k, Γ) -anonymity for

$$\Gamma = \{S : S \subseteq V, |S| \leq \ell, S \text{ is a } k'\text{-antiresolving set of } G \text{ with } k' < k\}.$$

According to this definition, it is sufficient to reduce the re-identification power of those vertex sets that originally are k' -antiresolving sets with $k' < k$, by enforcing the condition that these sets are k'' -antiresolving, with $k'' \geq k$, in the final graph. Note that $(k, \Gamma_{G, \ell})$ -anonymity is a conditional privacy property, in the sense that the privacy level of the perturbed graph is measured in terms of the initial threats present in the original graph. In particular, if a perturbed graph G' satisfies (k, ℓ) -anonymity, then it satisfies $(k, \Gamma_{G, \ell})$ -anonymity regardless of the original graph G . The converse is not true, as exemplified in Figure 3. In the figure, the sets $\{v_2\}$ and $\{v_3\}$ are 1-antiresolving sets of G_1 . Since both sets are 2-antiresolving sets of G_2 and G_3 , we have that G_2 and G_3 satisfy $(2, \Gamma_{G_1, 1})$ -anonymity. Moreover, the sets $\{v_1\}$ and $\{v_4\}$ are 2-antiresolving sets of G_2 , so this graph also satisfies $(2, 1)$ -anonymity. On the contrary, G_3 is not $(2, 1)$ -anonymous, as $\{v_1\}$ and $\{v_4\}$ are 1-antiresolving sets of this graph.

As we will show in Section 5, the notion of $(k, \Gamma_{G, \ell})$ -anonymity can be used to create variations of existing graph perturbation methods based on (k, ℓ) -anonymity that introduce a smaller number of modifications in the original graph, while providing the same level of protection against active attacks.

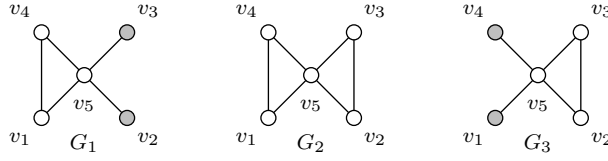


Fig. 3 The $(1,1)$ -anonymous graph G_1 and two graphs G_2 and G_3 satisfying $(2, \Gamma_{G_1,1})$ -anonymity. The graph G_2 also satisfies $(2,1)$ -anonymity, whereas G_3 does not.

3 Adjacency anonymity

In this section we continue analysing (k, ℓ) -anonymity in terms of the structural properties it requires on a graph. We observe that (k, ℓ) -anonymity assumes that the adversary knows every distance from the sybil nodes to other vertices of the graph. This makes it difficult for a graph to satisfy (k, ℓ) -anonymity even for small values of k or ℓ . For example, even-order cycles are regular and one may expect the users to be hidden behind the symmetry of the graph. However, as shown in [12], we have that no even-order cycle satisfies $(k, 1)$ -anonymity for any value of k greater than 1, because for any vertex u and its antipodal vertex v , we have $|\{v\}_{\{u\}}| = 1$, so $\{u\}$ is a 1-antiresolving set.

By examining current active attacks [1, 15, 16], we observe that none is able to handle arbitrary distances from the attacker nodes to the victims. In fact, they all rely on the neighbour relation rather than on shortest-paths. We argue that this is not only a characteristic of existing attacks, but rather a fundamental limitation of active adversaries. It seems unrealistic to expect that the adversary can keep control of arbitrary distance values. Accordingly, we propose (k, ℓ) -adjacency anonymity; a weaker, yet more realistic, adversarial model.

In a manner analogous to the definition of antiresolving sets, we use standard concepts from graph theory to represent an adversary whose knowledge about the victims consists of whether they are neighbours, or not, of each of the sybil nodes. The concept is known as *adjacency representation*, introduced by Jannesari and Omoomi [7], and defined as follows.

Definition 5 (Adjacency representation) Given a graph $G = (V, E)$, an ordered set $S = \{s_1, \dots, s_t\} \subseteq V$, and a vertex $u \in V$, the *adjacency representation* of u with respect to S is the vector $a_G(u \mid S) = (a_G(s_1, u), \dots, a_G(s_t, u))$ where the function $a_G: V \times V \rightarrow \mathbb{N}$ is defined by:

$$a_G(u, v) = \begin{cases} 0 & \text{if } u = v \\ 1 & \text{if } (u, v) \in E \\ 2 & \text{otherwise} \end{cases} \quad (1)$$

Note that $a_G(u, v) = \min\{2, d_G(u, v)\}$ for every $u, v \in V(G)$. Now, we will adapt the notion of a k -antiresolving set in order to account for the new type of adversary.

Definition 6 (k -adjacency antiresolving set) Let $G = (V, E)$ be a non-trivial graph. A set $S \subseteq V$ is a k -adjacency antiresolving set of G if k is the largest positive integer such that, for every $v \in V \setminus S$, there exist vertices $w_1, w_2, \dots, w_{k-1}$ such that $v, w_1, w_2, \dots, w_{k-1}$ are pairwise different and

$$a_G(v \mid S) = a_G(w_1 \mid S) = \dots = a_G(w_{k-1} \mid S).$$

To illustrate the difference between k -adjacency antiresolving sets and k -anti-resolving sets, consider again the case of even-order cycles discussed above. As we saw, every vertex u in an even-order cycle satisfies that $\{u\}$ is a 1-antiresolving set. However, if the order of the cycle is $n \geq 6$, then $\{u\}$ is a 2-adjacency antiresolving set, since it has exactly 2 neighbours and $n - 3 > 2$ non-neighbours. In order to measure a graph's increased resistance to an adversary that leverages adjacency representations, we enunciate the notions of (k, ℓ) -adjacency anonymity and $(k, \Gamma_{G, \ell})$ -adjacency anonymity, which relax the notions of (k, ℓ) -anonymity and $(k, \Gamma_{G, \ell})$ -anonymity, respectively. As above, we enunciate both properties as instances of a parametrisable notion called (k, Γ) -adjacency anonymity.

Definition 7 ((k, Γ)-adjacency anonymity) Let G be a non-trivial graph and let $\Gamma = \{S_1, S_2, \dots, S_t\}$, $S_i \subseteq V(G)$ for $i \in \{1, \dots, t\}$, be a family of subsets of $V(G)$. The graph G satisfies (k, Γ) -adjacency anonymity if every $S \in \Gamma$ is a k' -adjacency antiresolving set of G , with $k' \geq k$.

Definition 8 ((k, ℓ)-adjacency anonymity) A graph G is said to satisfy (k, ℓ) -adjacency anonymity if it satisfies (k, Γ) -adjacency anonymity for $\Gamma = \{S : S \subseteq V(G), |S| \leq \ell\}$.

At a conceptual level, (k, ℓ) -adjacency anonymity has some aspects in common with k -neighbourhood anonymity [23], as both consider an adversary provided with a neighbourhood relation. Adjacency anonymity, however, requires indistinguishability with respect to several specific subset of vertices, while such constraint is not imposed by k -neighbourhood anonymity.

Definition 9 (($k, \Gamma_{G, \ell}$)-adjacency anonymity) Let $G = (V, E)$ be a non-trivial graph and let $G' = (V, E')$ be the result of a series of perturbations on G . The graph G' is said to satisfy $(k, \Gamma_{G, \ell})$ -adjacency anonymity if it satisfies (k, Γ) -adjacency anonymity for

$$\Gamma = \{S : S \subseteq V, |S| \leq \ell, S \text{ is a } k'\text{-adjacency antiresolving set of } G \text{ with } k' < k\}.$$

The notion of $(k, \Gamma_{G, \ell})$ -adjacency anonymity will be the basis for the anonymisation algorithm that we will introduce in the next section.

4 An anonymisation method based on $(k, \Gamma_{G, 1})$ -adjacency anonymity

In order to showcase the usefulness of the new privacy properties, in this section we present a method for enforcing $(k, \Gamma_{G, 1})$ -adjacency anonymity, which is based on edge additions and removals. We highlight two features of this method. Firstly, it allows to increase the privacy of a graph G of order n up to satisfying $(k, \Gamma_{G, 1})$ -adjacency anonymity for any $k \in [2, \lfloor \frac{n-1}{2} \rfloor]$. Secondly, this privacy level is achieved with a minimum number of graph edits.

4.1 General notation

Before proceeding, we introduce some notation that will be used throughout this section. In a graph G , the *open neighbourhood* of a vertex $u \in V(G)$, denoted

by $N_G(u)$, is the set $N_G(u) = \{v \in V(G) \mid (u, v) \in E(G)\}$, whereas the *closed neighbourhood* of u in G is the set $N_G[u] = \{u\} \cup N_G(u)$. Similarly, for a set $S \subseteq V(G)$, we define $N_G(S) = \cup_{v \in S} N_G(v) \setminus S$ and $N_G[S] = \cup_{v \in S} N_G[v]$. The *degree* of a vertex u , denoted by $\delta_G(u)$, is its number of neighbours, i.e. $\delta_G(u) = |N_G(u)|$. In a graph G of order n , we will refer to vertices of degree 0 and $n-1$ as *isolated* and *dominant* vertices, respectively. For a graph $G = (V, E)$ and a subset S of vertices of G , we will denote by $\langle S \rangle_G$ the subgraph of G induced by S , that is $\langle S \rangle_G = (S, E \cap (S \times S))$. If there is no ambiguity, we will drop the graph-specific subindices and simply write $N(u)$, $\delta(u)$, etc. Also recall that, for a graph $G = (V, E)$ and a set $S \subseteq V$, $R_{G,S}$ is the equivalence relation such that two vertices u and v satisfy $(u, v) \in R_{G,S}$ if and only if $u, v \in V \setminus S$ and $a_G(u \mid S) = a_G(v \mid S)$. Moreover, we will use the notation $\mathcal{A}_{G,S}$ for the set of equivalence classes induced in $V \setminus S$ by the relation $R_{G,S}$. It is simple to see that S is a $(\min_{A \in \mathcal{A}_{G,S}} |A|)$ -adjacency antiresolving set of G .

4.2 The anonymisation method

Consider a $(k_0, 1)$ -adjacency anonymous graph G of order n . We will analyse the modifications necessary to turn G into a $(k, \Gamma_{G,1})$ -adjacency anonymous graph G' , with $k > k_0$. First, the following result characterises the values of k for which such a transformation may be of interest.

Proposition 1 *Let G be a non-complete, non-empty graph of order n satisfying $(k, 1)$ -adjacency anonymity. Then, $k \leq \lfloor \frac{n-1}{2} \rfloor$.*

Proof Let $G = (V, E)$ be a non-complete, non-empty graph of order n satisfying $(k, 1)$ -adjacency anonymity. Suppose, for the purpose of contradiction, that $k > \lfloor \frac{n-1}{2} \rfloor$. Let $v \in V$ be a vertex of G satisfying $\delta(v) \notin \{0, n-1\}$. The existence of such a vertex is guaranteed by the fact that the graph is neither complete nor empty. We have that $\mathcal{A}_{G,\{v\}} = \{N_G(v), V \setminus N_G[v]\}$. If $\delta(v) \leq \lfloor \frac{n-1}{2} \rfloor$, then $\{v\}$ is a k' -adjacency antiresolving set of G with $k' < k$, which is a contradiction. On the other hand, if $\delta(v) > \lfloor \frac{n-1}{2} \rfloor$, then $|V \setminus N_G[v]| \leq \lfloor \frac{n-1}{2} \rfloor$, which again means that $\{v\}$ is a k' -adjacency antiresolving set of G with $k' < k$, a contradiction. Therefore, we have that $k \leq \lfloor \frac{n-1}{2} \rfloor$. $\square$

In light of Proposition 1, aiming for values of k above $\lfloor \frac{n-1}{2} \rfloor$ is of little practical interest, since complete and empty graphs are not useful for analysis. Thus, in what follows we will focus on the values of k in the interval $[2, \lfloor \frac{n-1}{2} \rfloor]$. Now, the following result enunciates the conditions that must be enforced in order to transform a graph $G = (V, E)$ into a graph $G' = (V, E')$ satisfying $(k, \Gamma_{G,1})$ -adjacency anonymity for some lower-bounded value of k .

Theorem 1 *Let $G = (V, E)$ be a non-trivial graph and let $G' = (V, E')$ be a graph obtained from G by edge additions and removals. Let k be a positive integer satisfying $1 \leq k \leq |V|-1$. Then the graph G' satisfies $(k', \Gamma_{G,1})$ -adjacency anonymity with $k' \geq k$ if and only if every vertex $v \in V$ such that $1 \leq \delta_G(v) < k$ or $|V|-k-1 < \delta_G(v) \leq |V|-2$ satisfies one of the following conditions:*

- (i) $\delta_{G'}(v) = 0$
- (ii) $\delta_{G'}(v) = |V| - 1$

(iii) $k \leq \delta_{G'}(v) \leq |V| - k - 1$

Proof Let $v \in V$ be a vertex such that $\{v\}$ is a k' -adjacency antiresolving set of G with $k' < k$. If $\delta_{G'}(v) = 0$ or $\delta_{G'}(v) = |V| - 1$, we have that $\mathcal{A}_{G',\{v\}} = \{V \setminus \{v\}\}$, so the set $\{v\}$ is a $(|V| - 1)$ -adjacency antiresolving set of G' . Recall that, according to the premises of the theorem, $|V| - 1 \geq k$. Now we address the case where $k \leq \delta_{G'}(v) \leq |V| - k - 1$. Here, we have that $\mathcal{A}_{G',\{v\}} = \{N_{G'}(v), V \setminus N_{G'}[v]\}$, so the set $\{v\}$ is a $(\min\{\delta_{G'}(v), |V| - \delta_{G'}(v) - 1\})$ -adjacency antiresolving set of G' . Since $\delta_{G'}(v) \geq k$ and $|V| - \delta_{G'}(v) - 1 \geq k$, we have that $\{v\}$ is a k'' -antiresolving set of G' with $k'' \geq k$. In conclusion, G' satisfies $(k', \Gamma_{G,1})$ -adjacency anonymity with $k' \geq k$.

To conclude our proof, let us now assume that there exists a vertex $v \in V$ such that $\{v\}$ is a k' -adjacency antiresolving set of G with $k' < k$ and either $1 \leq \delta_{G'}(v) < k$ or $|V| - k - 1 < \delta_{G'}(v) \leq |V| - 2$. In both cases, we have that $\mathcal{A}_{G',\{v\}} = \{N_{G'}(v), V \setminus N_{G'}[v]\}$. Since either $|N_{G'}(v)| < k$ or $|V \setminus N_{G'}[v]| < k$, we have that G' does not satisfy $(k'', \Gamma_{G,1})$ -adjacency anonymity for any $k'' \geq k$. $\square$

Our anonymisation method is based on Theorem 1. It receives as input a $(k_0, 1)$ -adjacency anonymous graph $G = (V, E)$ and an integer k such that $k_0 < k \leq \left\lfloor \frac{|V|-1}{2} \right\rfloor$, and efficiently obtains a $(k, \Gamma_{G,1})$ -adjacency anonymous graph $G' = (V, E')$. As we mentioned above, the method works by performing a series of edge additions and removals upon G , as outlined in Algorithm 1 (EDIT-GRAPH).

Algorithm 1 Given a $(k_0, 1)$ -adjacency anonymous graph $G = (V, E)$ and an integer $k \in [k_0 + 1, \left\lfloor \frac{|V|-1}{2} \right\rfloor]$, EDIT-GRAPH provides a $(k, \Gamma_{G,1})$ -adjacency anonymous graph $G' = (V, E')$.

```

1: procedure EDIT-GRAPH( $G = (V, E), k$ )
2:   Sort the elements of  $V$  by degree
3:    $E' \leftarrow E$   $\triangleright G' = (V, E')$ 
4:    $L_0 \leftarrow \{v \in V : 1 \leq \delta_G(v) < k\}$ 
5:    $H_0 \leftarrow \{v \in V : |V| - k - 1 < \delta_G(v) \leq |V| - 2\}$ 
6:    $L \leftarrow L_0$   $\triangleright$  The original value will be used later
7:   while  $L \neq \emptyset$  do
8:     Pick a pair  $u, v \in L$  s.t.  $(u, v) \notin E'$ 
9:     if no such pair exists then
10:       Pick  $u \in L, v \in V \setminus L$  s.t.  $(u, v) \notin E'$ 
11:        $E' \leftarrow E' \cup \{(u, v)\}$ 
12:        $L \leftarrow \{v \in L : 1 \leq \delta_{G'}(v) < k\}$   $\triangleright$  Update  $L$ 
13:    $H \leftarrow \{v \in H_0 : |V| - k - 1 < \delta_{G'}(v) \leq |V| - 2\}$ 
14:   while  $H \neq \emptyset$  do
15:     Pick a pair  $u, v \in H$  s.t.  $(u, v) \in E'$ 
16:     if no such pair exists then
17:       Pick  $u \in H, v \in V \setminus (H \cup L_0)$  s.t.  $(u, v) \in E'$ 
18:        $E' \leftarrow E' \setminus \{(u, v)\}$ 
19:        $H \leftarrow \{v \in H : |V| - k - 1 < \delta_{G'}(v) \leq |V| - 2\}$   $\triangleright$  Update  $H$ 
20: return  $G'$ 

```

In Algorithm EDIT-GRAPH, the sets L and H are initialised with the vertices whose degrees in G are, respectively, smaller and greater than required for the

privacy requirement to be satisfied (without being isolated nor dominant vertices). Then, by adding or removing edges, these vertices are forced to satisfy condition (iii) of Theorem 1. Note that, at step 12, when updating L , if the degree of an originally isolated vertex x has increased as a result of some edge addition, it is not included in L . This is so even if $1 \leq \delta_{G'}(x) < k$, because $\{x\} \notin \Gamma_{G,1}$. An analogous criterion is applied at step 19 to any originally dominant vertex if its degree decreases after an edge removal. The rationale behind the loop in steps 7 to 12 is to first add as many edges as possible between pairs of vertices from L , since every addition of this type contributes to both vertices getting closer to satisfy the required condition. When such additions are no longer possible, we then add edges linking a vertex $u \in L$ and a vertex $v \notin L$ whose degree is as small as possible, so the degree of vertices from H is only increased if there is no vertex in $V \setminus H$ to which u can be linked. An analogous idea is applied in steps 14 to 19 to first remove edges joining pairs of vertices from H , then edges joining a vertex from H to another vertex, with the particularity that step 17 takes care of not making the degree of a vertex in L_0 decrease again. For the values of k that we are considering, at least the edge addition and removal described in steps 10 and 17, respectively, are always possible. These operations are efficiently performed by maintaining the elements of V sorted by their degree and updating the ordering when necessary.

4.2.1 Complexity and optimality analysis

We first analyse the number of modifications performed by the algorithm. The best scenario occurs when all edge additions are done according to step 8, and all edge removals are done according to step 15, as shown in the following results.

Theorem 2 *Let $G = (V, E)$ be a $(k_0, 1)$ -adjacency anonymous social graph and let $k \in [k_0 + 1, \lfloor \frac{|V|-1}{2} \rfloor]$. The number t of edges added by steps 7 to 12 of Algorithm EDIT-GRAPH satisfies*

$$\left\lceil \frac{1}{2} \times \sum_{\substack{u \in V, \\ 1 \leq \delta_G(u) < k}} k - \delta_G(u) \right\rceil \leq t \leq \sum_{\substack{u \in V, \\ 1 \leq \delta_G(u) < k}} k - \delta_G(u) \quad (2)$$

Proof Due to its length, we develop this proof in Appendix A. $\square$

The lower and upper bounds provided in Theorem 2 are tight, as exemplified in Figures 4.a) and 4.b), respectively.

The next result describes the number t' of edges removed by steps 14 to 19 of Algorithm EDIT-GRAPH.

Theorem 3 *Let $G = (V, E)$ be a $(k_0, 1)$ -adjacency anonymous social graph and let $k \in [k_0 + 1, \lfloor \frac{|V|-1}{2} \rfloor]$. Let G_t be the graph obtained from G after executing steps 7 to 12 of Algorithm EDIT-GRAPH. The number t' of edges removed by steps 14 to 19 of Algorithm EDIT-GRAPH satisfies*

$$t' \geq \left\lceil \frac{1}{2} \times \sum_{\substack{u \in V, \\ |V|-k-1 < \delta_G(u) \leq |V|-2}} [k - (|V| - \delta_{G_t}(u) - 1)] \right\rceil \quad (3)$$

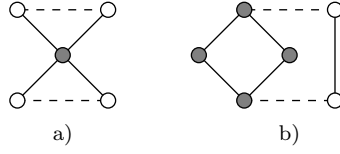


Fig. 4 Two examples where the number of edges added by steps 7 to 12 of Algorithm EDIT-GRAPH (for $k = 2$) reaches the (a) lower and (b) upper bounds of the inequalities in Equation 2. In both cases, dashed lines indicate the edges added by the algorithm.

and

$$t' \leq \sum_{\substack{u \in V \\ |V| - k - 1 < \delta_G(u) \leq |V| - 2}} [k - (|V| - \delta_{G_t}(u) - 1)] \quad (4)$$

Proof As in the previous case, we develop this proof in Appendix B. $\square$

The worst-case computational complexity of Algorithm EDIT-GRAPH on a graph G of order n is $\mathcal{O}(n^2)$, because every pair of vertices $u, v \in V(G) \times V(G)$, $u \neq v$, is susceptible of being evaluated once, either as a possible edge addition in steps 7 to 12, or as a possible edge removal in steps 14 to 19. However, in practice, especially for low density graphs, steps 7 to 19 are performed in quasi-linear time, so the entire process is dominated by the $\mathcal{O}(n \log n)$ complexity of step 2. This observation is relevant in view of the fact that real-life social graphs are characterised by having very low density. Additionally, in this scenario, and for the values of k that we are considering, steps 14 to 19 are very unlikely to be executed at all, because no vertex has a considerably large degree with respect to the number of vertices of the graph.

Summarising. We have introduced in this section an anonymisation technique (Algorithm EDIT-GRAPH) to transform social graphs into $(k, \Gamma_{G,1})$ -adjacency anonymous graphs for values of k up to $\lfloor \frac{n-1}{2} \rfloor$, where n is the order of the graph. The proposed method is efficient and has theoretical tight bounds on the number of graph modifications it performs. Evaluating the actual perturbation of Algorithm EDIT-GRAPH and the privacy offered by the resulting perturbed graphs is the aim of the next section.

5 Experiments

The purpose of the experiments reported in this section is to assess several aspects of our new proposals. First, we compare the effect of enforcing $(k, \Gamma_{G,\ell})$ -anonymity to that of enforcing (k, ℓ) -anonymity, in terms of privacy preservation and the number of graph edits performed. Additionally, we compare the methods based on these privacy notions to our Algorithm EDIT-GRAPH, which is based on the notion of $(k, \Gamma_{G,\ell})$ -adjacency anonymity, and assess the effect of the value of k on the number of graph edits performed by Algorithm EDIT-GRAPH. In these experiments, we consider an active attacker as the one evaluated in [12], who has the ability to insert one sybil node into the graph. Then, to conclude our study, we assess the extent to which Algorithm EDIT-GRAPH is able to provide protection from

active attackers leveraging sets of of sybil nodes larger than those covered by the theoretical privacy guarantee.

5.1 Experimental setup

The evaluation is conducted on a large collection of randomly generated graphs. This collection is composed of 9.7 million graphs: 100,000 for each density value in the set $\{0.03, 0.04, \dots, 1\}$ ². Each graph has 200 vertices, and its edge set is randomly generated in such a manner that the desired density value is met.

The chosen attack is the *walk-based attack* proposed in [1]. Let $S = \{x_1, \dots, x_n\}$ be the set of sybil nodes enrolled by the adversary and let $T = \{y_1, \dots, y_m\}$ be the set of victims. The walk-based attack creates a unique fingerprint for each victim $y_i \in T$ by randomly selecting a subset $N_i \subseteq S$ such that $N_i \neq N_j$ for every pair of different $i, j \in \{1, \dots, m\}$, and connecting y_i to every $x \in N_i$. In order to make the subgraph induced by the set of sybil nodes uniquely retrievable with high probability, the attack first adds all edges (x_i, x_{i+1}) , $1 \leq i \leq n-1$, and then every other pair of sybil nodes is linked by an edge with probability $\frac{1}{2}$.

Let $G = (V, E)$ be a graph from the collection, $G' = (V \cup S, E \cup E')$, with $E' \subseteq S \times (V \cup S)$, the graph obtained after simulating the action of the attacker, and $G'' = (V(G'), (E(G') \cup A) \setminus B)$, with $A \subseteq (V(G') \times V(G') \setminus E(G'))$ and $B \subseteq E(G')$, the result of applying a perturbation method on G' . The probability of success for the walk-based attack is computed by the following formula [12]:

$$\Pr = \begin{cases} \frac{\sum_{X \in \mathcal{X}} \prod_{1 \leq i \leq m} p_{i,X}}{|\mathcal{X}|} & \text{if } \mathcal{X} \neq \emptyset \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $\mathcal{X}$ contains all ordered subsets X of $V(G')$ such that $\langle X \rangle_{G''} \cong \langle S \rangle_{G'}$, $V_{i,X}$ contains all vertices $v \in V(G')$ that satisfy $a_{G''}(v \mid X) = a_{G'}(y_i \mid S)$, and

$$p_{i,X} = \begin{cases} \frac{1}{|V_{i,X}|} & \text{if } y_i \in V_{i,X} \\ 0 & \text{otherwise.} \end{cases}$$

5.2 Comparison of the privacy properties

We compare three perturbation-based social graph anonymisation methods, one of which is based on the original notion of (k, ℓ) -anonymity and the other two on the new privacy notions introduced in this paper: $(k, \Gamma_{G,\ell})$ -anonymity and $(k, \Gamma_{G,\ell})$ -adjacency anonymity.

The first method was introduced by Mauw et al. in [12]. It transforms a $(1, 1)$ -anonymous graph into a graph that satisfies (k, ℓ) -anonymity for some $k > 1$ or some $\ell > 1$. This method works by iteratively adding edges to the original graph. At each step, the method finds a 1-antiresolving set $\{v\}$ and modifies the graph by adding an edge that induces an odd-order cycle in an eccentricity path of v . The

² Although it is not a requirement of Algorithm EDIT-GRAPH, the other two methods that we evaluate in these experiments require the original graph to be connected. We start with the density value 0.03 because there exist no connected graphs of order 200 and density 0.02 or 0.01.

method ends when no unitary 1-antiresolving sets are found. As discussed in [12], a problem faced by this method is that an edge addition may cause other unitary vertex sets to become 1-antiresolving, which in turn causes the algorithm to add more edges to ensure that the privacy guarantee is satisfied.

The second method is a variation on the previous one, aiming to transform a $(1, 1)$ -anonymous graph G into a graph G' satisfying $(k, \Gamma_{G,1})$ -anonymity for some $k > 1$. The modified method differs from the original method given in [12] in two aspects:

1. At each iteration step, a set $\{v\}$ is selected such that it is 1-antiresolving in both the current and the original graph. In the original method, $\{v\}$ is required to be a 1-antiresolving set of the current graph, even if it was k -antiresolving, with $k > 1$, in the original graph.
2. The iteration stops when every 1-antiresolving set $\{v\}$ of the original graph is k -antiresolving in the current graph with $k > 1$. In the original method, the iteration stops when the current graph has no unitary 1-antiresolving sets, including those that were originally k -antiresolving with $k > 1$.

Finally, the third method in consideration is our algorithm EDIT-GRAPH, with $k = 2$, which guarantees the perturbed graph to be $(2, \Gamma_{G,1})$ -adjacency anonymous. We ran the three methods on the graph collection described above. A walk-based attack leveraging one sybil node is simulated on each graph, and for each combination of a perturbation method and a density value, we computed the average of the values obtained for the success probability of the attack and the number of modifications performed, over the corresponding 100,000 graphs. For the first two methods, the number of modifications is the number of added edges, whereas for algorithm EDIT-GRAPH it is the sum of the number of added edges (steps 7–12) and the number of removed edges (steps 14–19).

The first relevant result from these runs is that, after perturbing the graph with each of the three methods, the success probability of the walk-based attack with one sybil node is zero for all density values. This fact had already been reported for the first method in [12]. Here, we highlight the fact that the other two methods display the same behaviour, which shows that the notions of conditional privacy introduced by $(k, \Gamma_{G,\ell})$ -anonymity and $(k, \Gamma_{G,\ell})$ -adjacency anonymity are able to provide a level of protection against the walk-based attack (or similar neighbourhood-based attacks) that is at least as good as the one provided by (k, ℓ) -anonymity.

The advantages of using the new privacy properties become clear when we compare the average number of modifications introduced on the original graphs by each method, as depicted in Figure 5. As can be observed in the figure, the number of modifications performed by the method based on $(k, \Gamma_{G,\ell})$ -anonymity is at most the same performed by the method based on (k, ℓ) -anonymity. Moreover, for several density values, it is up to five times smaller. It is worth noting that the largest differences occur for small density values, which are typically observed in real-world social graphs. In our opinion, the most important conclusion that can be extracted from Figure 5 is that adjusting the adversary model to a more realistic view of the capabilities of active adversaries allows for a considerable reduction of the number of necessary perturbations. This is due to the fact that the privacy notions based on the metric representation tend to “over-protect” vertices that appear as re-identifiable, but really are not. Methods based on $(k, \Gamma_{G,\ell})$ -

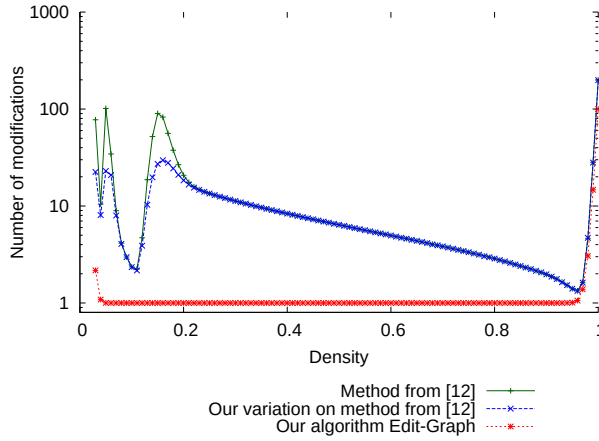


Fig. 5 Number of modifications done by the methods enforcing (k, ℓ) -anonymity for some $k > 1$ or some $\ell > 1$ [12], $(k, \Gamma_{G,1})$ -anonymity with $k > 1$ (our variation on the method from [12]) and $(2, \Gamma_{G,1})$ -adjacency anonymity (Alg. EDIT-GRAPH with $k = 2$).

adjacency anonymity do not take these unnecessary measures, thus altering the graph structure considerably less, as seen in the figure.

5.3 The effect of k in Algorithm EDIT-GRAPH

In the previous section we saw the number of modifications introduced by our Algorithm EDIT-GRAPH for $k = 2$, as this is sufficient for comparing this algorithm with its two counterparts. Here, we go into more detail regarding the effect of rising the value of k on the number of modifications performed by the algorithm. To that end, Figure 6 shows the number of modifications performed for $k \in \{2, \dots, 8\}$.

The fact that larger values of k require the algorithm to perform more graph modifications is a direct consequence of the considered privacy property, so we will focus on other interesting facts that can be observed in the figure. The smallest number of modifications are performed for intermediate density values. The reason for this behaviour is the fact that, in these cases, almost no vertex corresponding to a legitimate user (that is, a non-sybil node) has a degree value in the intervals specified in Theorem 1, so the only modifications necessary tend to be the $k - 1$ edge additions necessary for rising the degree of the inserted sybil node to k . This is not the case for the smallest density values, where more edge additions may be necessary; and the largest density values, where edge removals are necessary to lower the degree of some vertices, whereas edge additions are necessary to rise the degree of the inserted sybil node.

To conclude, it is worth remarking that, in all the scenarios depicted in Figure 6, the success probability of the walk-based attack with one sybil node continues to be zero.

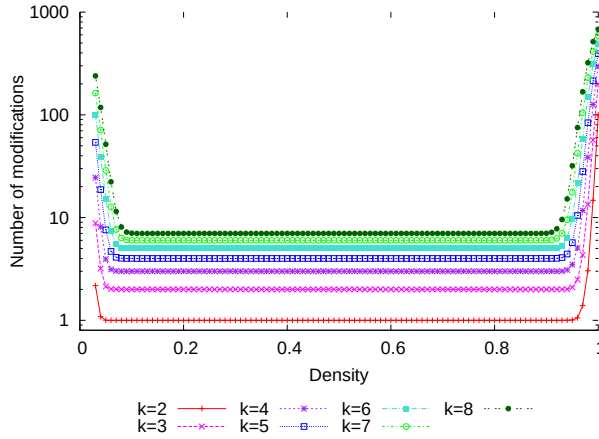


Fig. 6 Number of modifications performed by Algorithm EDIT-GRAPH for several values of k .

5.4 Protection against attackers leveraging larger amounts of sybil nodes

It was pointed out in [12] that graph perturbation methods aiming to enforce (k, ℓ) -anonymity for some $k > 1$ and $\ell = 1$ are to some extent able to thwart active attacks leveraging more than one sybil node. This was due to the fact that, by altering the subgraph induced by the sybil nodes and/or the fingerprints of a subset of victims, the adversary is prevented from retrieving the set of sybil nodes, or correctly re-identifying the victims.

Here, we analyse to what extent Algorithm EDIT-GRAPH shows an analogous behaviour. To that end, we simulated active attacks leveraging up to eight sybil nodes on the graphs of the collection with density values $\{0.1, 0.2, \dots, 1\}$. Recall that, according to [1], eight sybil nodes are sufficient for building a uniquely retrievable subgraph and re-identifying all vertices of a graph of order 200, as the ones in our collection. Then, seven runs of Algorithm EDIT-GRAPH, with $k \in \{2, \dots, 8\}$, were made on each resulting graph. Figure 7 shows the results of these runs using colour matrices. The colour of each cell represents the averaged success probabilities of the attacks on the perturbed graphs. For each cell's RGB colour value, $R = 255 \cdot \text{Pr}$, $G = 0$ and $B = 255 \cdot (1 - \text{Pr})$, where Pr stands for the success probability of the attacks; that is, red cells (or cells with a prevailing red component) represent highly successful attacks, and prevailingly blue cells represent thwarted attacks.

From the analysis of the figure, the most important observation is that starting at values of k close to the number of sybil nodes, the success probability of the walk-based attack rapidly falls from values close to 1 to values close to 0. In other words, by increasing the value of k , a point is reached where attacks start to be successfully thwarted. This is a consequence of the manner in which edges between sybil nodes are generated in the attack, which makes all sybil nodes have very similar degrees. Thus, depending on the value of k , either the algorithm modifies almost no sybil node's degree, or it modifies almost all, in which case the attacker fails to retrieve the sybil subgraph. Moreover, this behaviour is negligibly affected by the original graph's density, as illustrated by the considerably large similarity

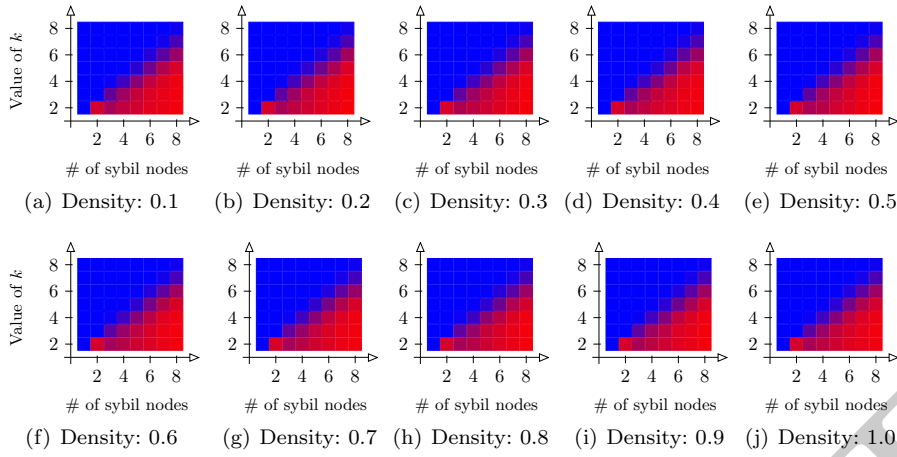


Fig. 7 Success probabilities of the walk-based attack, inserting 1 to 8 sybil nodes, on randomly generated graph collections for densities 0.1, 0.2, ..., 1. For each graph, Algorithm EDIT-GRAPH was run with $k \in \{2, \dots, 8\}$. For each cell's RGB colour value, $R = 255 \cdot \text{Pr}$, $G = 0$ and $B = 255 \cdot (1 - \text{Pr})$, where Pr stands for the success probability of the attacks.

among the colour matrices (a) to (j) in the figure, because the density of the attacker subgraph (always slightly above 0.5) is determined by the specification of the attack, not by the density of the original graph.

5.5 Summary of experimental results

The experiments described in this section show that, when compared to (k, ℓ) -anonymity, the new notions of $(k, \Gamma_{G, \ell})$ -anonymity and $(k, \Gamma_{G, \ell})$ -adjacency anonymity allow to provide protection from equally capable active attackers while introducing a much smaller number of changes in the graph structure. Moreover, although the theoretical privacy guarantee offered by Algorithm EDIT-GRAPH ($(k, \Gamma_{G, 1})$ -adjacency anonymity) only concerns attackers with the ability to insert one sybil node in the network, the algorithm is also capable of thwarting attacks from more capable adversaries.

6 Related work

Several graph-oriented notions of k -anonymity have been proposed. As we mentioned above, Liu and Terzi [9] considered an adversary who knows the degree of the victim node. They defined the notion of k -degree anonymity, which is satisfied if for every vertex of the graph there exist other $k - 1$ vertices with the same degree. Liu and Terzi devised a simple and efficient algorithm to transform a graph into a k -degree anonymous graph. The authors also show that the utility of the graph is at least partly kept, because the average path length and the exponent of the power-law distribution of the original graph are preserved. Further improvements

of this type of methods, aiming for increased utility levels, have been proposed [10, 2, 5, 11, 17].

A privacy notion that is strictly stronger than k -degree anonymity is k -neighbourhood anonymity [23]. This property requires that for every vertex v in the graph there exist at least $k-1$ other vertices $v_1, \dots, v_{k-1}$ such that the subgraph induced by v 's neighbours is isomorphic to the subgraph induced by v_i 's neighbours, for every $i \in \{1, \dots, k-1\}$. It is simple to see that every k -neighbourhood anonymous graph is k' -degree anonymous for some $k' \geq k$. Likewise, a strictly stronger privacy notion is that of k -automorphism anonymity [24]. A graph G is said to be k -automorphic if there exist $k-1$ different automorphisms $\varphi_1, \dots, \varphi_{k-1}$ of G such that $\varphi_i(v) \neq \varphi_j(v)$ for every $v \in V(G)$ and every pair φ_i, φ_j , $1 \leq i < j \leq k-1$. Alternative formulations of this privacy notion were presented independently as k -symmetry [21] and k -isomorphism [4]. A particularity of the latter is that it forces the creation of a disconnected graph with k connected components, which are pairwise isomorphic. A natural trade-off between the strength of the privacy notions and the amount of structural disruption caused by the anonymisation methods based on them has been empirically demonstrated in [24].

As we mentioned in previous sections, Backstrom et al. were the first to show the impact of active attacks in social networks [1]. They introduced the attack methodology, of which the walk-based attack used in Section 5 is an instance, consisting in planting a uniquely identifiable and efficiently retrievable attacker subgraph and creating unique fingerprints for the victims determined by their links to the attacker subgraph. A combination of active and passive strategies was introduced by Peng et al. [15, 20]. Their attack relies on a small set of sybil nodes to re-identify an initial set of victims, and then uses the information from an auxiliary, non-anonymised graph, to iteratively re-identify neighbours of previously re-identified victims.

To the best of our knowledge, (k, ℓ) -anonymity [19] has been until now the only privacy property tailored to measure the resistance of social graphs to active attacks. Likewise, a single algorithm, that of Mauw et al. [12], has been proposed specifically for protecting graphs from active attacks. This method uses the notion of (k, ℓ) -anonymity, but does not guarantee anonymity for k larger than 2, and in doing so it may add a proportionally large number of edges to the original graph. We remedy both issues in this article.

The work presented in this paper, along with those mentioned above, falls into one of the two main approaches of privacy-preserving social graph publication, namely that of publishing a new, anonymised version of the original graph. The other approach consists in generating a randomised model of the graph and using it to answer queries about the graph structure. For example, Hay et al. [6] create a new, *generalised* graph where every vertex represents at least k vertices of the original graph. This graph is complete, and every generalised edge is weighted with the probability of an edge existing (in the original graph) between a pair of vertices randomly taken from each generalised vertex. Whenever a query is posed, they sample a graph from the generalised graph and use it to answer the query. The main problem with this approach, pointed out in [24], is that sampled graphs can dramatically differ from each other, even when repeatedly answering the same query. A somewhat related approach is presented in [13, 8]. Instead of creating a generalised graph, they create a new graph with the same vertex set and a randomly generated edge set. This graph is then used for answering neighbourhood

queries. A fundamental difference between the work in [13,8] and the ones previously discussed in this paper, is that the former addresses a setting where vertex ids are public and only the existence of edges between vertices has to be kept private. In randomisation-based approaches, privacy is generally measured in terms of the increase in the adversary’s inference power after releasing each chunk of data. For example, in [13,8], they measure the difference between the probabilities of the adversary correctly determining the existence of a set of edges before and after a (noisy) neighbourhood query is answered.

7 Concluding remarks

We have re-assessed the notion of (k, ℓ) -anonymity, which quantifies the privacy level of a social graph in the presence of active adversaries. Firstly, we have introduced the notion of $(k, \Gamma_{G, \ell})$ -anonymity, which alleviates the computational cost of using (k, ℓ) -anonymity as the basis of anonymisation methods based on edge set perturbations. The new privacy property also allows us to reduce the amount of perturbation needed to protect a social graph from an active attack. Secondly, we have critically assessed one of the assumptions posed by (k, ℓ) -anonymity on the adversary capabilities. Judging that it is unrealistic to assume that an adversary will be able to know all distances between a set of sybil nodes and every other vertex of the social graph, we introduced the notion of adjacency anonymity, which accounts for adversaries who control the connection patterns with the neighbours of the sybil nodes. Finally, combining the two previous ideas, we have introduced a new privacy property: $(k, \Gamma_{G, \ell})$ -adjacency anonymity. Based on this new property, we proposed an efficient algorithm for transforming a graph G into a $(k, \Gamma_{G, 1})$ -adjacency anonymous graph, for values of k up to $\lfloor \frac{n-1}{2} \rfloor$, where n is the number of vertices of the graph. We have additionally determined tight bounds on the number of edits performed by this method. We conducted a series of experiments on a collection of randomly generated graphs, which show that, when compared to (k, ℓ) -anonymity, the new privacy notions continue to provide protection from equally capable active attackers while requiring a much smaller number of graph perturbations.

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A Proof of Theorem 2 (See Page 13)

Theorem 2 Let $G = (V, E)$ be a $(k_0, 1)$ -adjacency anonymous social graph and let $k \in [k_0 + 1, \lfloor \frac{|V|-1}{2} \rfloor]$. The number t of edges added by steps 7 to 12 of Algorithm EDIT-GRAPH satisfies

$$\left\lceil \frac{1}{2} \times \sum_{\substack{u \in V, \\ 1 \leq \delta_G(u) < k}} k - \delta_G(u) \right\rceil \leq t \leq \sum_{\substack{u \in V, \\ 1 \leq \delta_G(u) < k}} k - \delta_G(u) \quad (2)$$

Proof Let $((u_1, v_1), (u_2, v_2), \dots, (u_t, v_t))$, with $(u_i, v_i) \in (V \times V) \setminus E$ for $i \in \{1, \dots, t\}$, be the sequence of edges added to G by steps 7 to 12 of Algorithm EDIT-GRAPH. Let $E_0 = E$ and $E_i = E_{i-1} \cup \{(u_i, v_i)\}$, for $i \in \{1, \dots, t\}$. Moreover, for every $i \in \{0, \dots, t\}$, let $G_i = (V, E_i)$ and $L_i = \{v \in V : 1 \leq \delta_{G_i}(v) < k\}$.

After adding the edge (u_i, v_i) , we have that $\delta_{G_i}(u_i) = \delta_{G_{i-1}}(u_i) + 1$ and $\delta_{G_i}(v_i) = \delta_{G_{i-1}}(v_i) + 1$, whereas $\delta_{G_i}(x) = \delta_{G_{i-1}}(x)$ for every $x \in V - \{u_i, v_i\}$.

We define the function

$$\text{missing}(G_i) = \sum_{x \in L_i} (k - \delta_{G_i}(x))$$

which specifies by how much the sum of the degrees of vertices from L needs to be increased for G_i to satisfy $(k, I_{G,1})$ -adjacency anonymity. Note that, by the definition of t , we have that $\text{missing}(G_t) = 0$. Moreover, $\text{missing}(G_0) = \sum_{u \in V, 1 \leq \delta_G(u) < k} k - \delta_G(u)$. After adding the edge (u_i, v_i) , the following situations are possible:

1. $u_i, v_i \in L_{i-1}$. In this case, since two vertices from L_{i-1} have their degree increased by 1, we have that $\text{missing}(G_i) = \text{missing}(G_{i-1}) - 2$.
2. $u_i \in L_{i-1}$ and $v_i \notin L_{i-1}$, or *vice versa*. Here, $\text{missing}(G_i) = \text{missing}(G_{i-1}) - 1$.

With the previous definitions in mind, we will address the proof of the left-hand inequality in Equation 2. To that end, we will assume, for the purpose of contradiction, that

$$t < \left\lceil \frac{\sum_{u \in V, 1 \leq \delta_G(u) < k} k - \delta_G(u)}{2} \right\rceil = \left\lceil \frac{\text{missing}(G_0)}{2} \right\rceil.$$

If $\text{missing}(G_0)$ is even, we have that $t < \frac{\text{missing}(G_0)}{2}$. Given that, in the best case scenario, situation 1 above occurs at every iteration of the algorithm, we have

$$\begin{aligned} \text{missing}(G_t) &\geq \text{missing}(G_0) - 2t \\ &> \text{missing}(G_0) - 2 \cdot \frac{\text{missing}(G_0)}{2} \\ &= 0 \end{aligned}$$

which is a contradiction.

In a similar manner, if $\text{missing}(G_0)$ is odd, we have that $t < \frac{\text{missing}(G_0)+1}{2}$. Here, in the best case scenario, situation 1 above occurs in every iteration, except one, so

$$\begin{aligned} \text{missing}(G_t) &\geq \text{missing}(G_0) - 2(t-1) - 1 \\ &> \text{missing}(G_0) \\ &\quad - 2 \left(\frac{\text{missing}(G_0)+1}{2} - 1 \right) - 1 \\ &= 0 \end{aligned}$$

which is also a contradiction. Thus, we can conclude that

$$t \geq \left\lceil \frac{\sum_{u \in V, 1 \leq \delta_G(u) < k} k - \delta_G(u)}{2} \right\rceil.$$

The right-hand inequality in Equation 2 is trivial, given that at least one vertex has its degree increased by 1 at every iteration. The proof is thus complete. $\square$

B Proof of Theorem 3 (See Page 13)

Theorem 3 Let $G = (V, E)$ be a $(k_0, 1)$ -adjacency anonymous social graph and let $k \in [k_0 + 1, \lfloor \frac{|V|-1}{2} \rfloor]$. Let G_t be the graph obtained from G after executing steps 7 to 12 of

Algorithm EDIT-GRAPH. The number t' of edges removed by steps 14 to 19 of Algorithm EDIT-GRAPH satisfies

$$t' \geq \left\lceil \frac{1}{2} \times \sum_{\substack{u \in V, \\ |V|-k-1 < \delta_G(u) \leq |V|-2}} [k - (|V| - \delta_G(u) - 1)] \right\rceil \quad (3)$$

and

$$t' \leq \sum_{\substack{u \in V \\ |V|-k-1 < \delta_G(u) \leq |V|-2}} [k - (|V| - \delta_G(u) - 1)] \quad (4)$$

Proof We will follow a reasoning analogous to the one applied in the proof of Theorem 2. Let $((u_1, v_1), (u_2, v_2), \dots, (u_{t'}, v_{t'}))$, with $(u_i, v_i) \in E_t = E(G_t)$ for $i \in \{1, \dots, t'\}$, be the sequence of edges removed from G_t by steps 14 to 19 of Algorithm EDIT-GRAPH. Let $E_{t+i} = E_{t+i-1} \setminus \{(u_i, v_i)\}$, for $i \in \{1, \dots, t'\}$. Moreover, for every $i \in \{1, \dots, t'\}$, let $G_{t+i} = (V, E_{t+i})$ and $H_i = \{v \in V : |V| - k - 1 < \delta_{G_{t+i}}(v) \leq |V| - 2\}$.

After removing the edge (u_i, v_i) , we obtain that $\delta_{G_{t+i}}(u_i) = \delta_{G_{t+i-1}}(u_i) - 1$ and $\delta_{G_{t+i}}(v_i) = \delta_{G_{t+i-1}}(v_i) - 1$, whereas $\delta_{G_{t+i}}(x) = \delta_{G_{t+i-1}}(x)$ for every $x \in V - \{u_i, v_i\}$.

Now we introduce the function

$$\text{excess}(G_{t+i}) = \sum_{x \in H_i} [k - (|V| - \delta_{G_{t+i}}(x) - 1)].$$

In a manner analogous to the proof of Theorem 2, we have that by definition $\text{excess}(G_{t+t'}) = 0$ and

$$\text{excess}(G_t) = \sum_{u \in V, |V|-k-1 < \delta_G(u) < |V|-2} [k - (|V| - \delta_G(u) - 1)].$$

Additionally, after removing the edge (u_i, v_i) , the following situations are possible:

1. $u_i, v_i \in H_{i-1}$. In this case, since two vertices from H_{i-1} have their degree decreased by 1, we have that $\text{excess}(G_{t+i}) = \text{excess}(G_{t+i-1}) - 2$.
2. $u_i \in H_{i-1}$ and $v_i \notin H_{i-1}$, or *vice versa*. Here, $\text{excess}(G_{t+i}) = \text{excess}(G_{t+i-1}) - 1$.

Now, to address the proof of the inequality in Equation 3, we assume, for the purpose of contradiction, that

$$\begin{aligned} t' &< \left\lceil \frac{\sum_{u \in V, |V|-k-1 < \delta_G(u) \leq |V|-2} [k - (|V| - \delta_G(u) - 1)]}{2} \right\rceil \\ &= \left\lceil \frac{\text{excess}(G_t)}{2} \right\rceil. \end{aligned}$$

In consequence, if $\text{excess}(G_t)$ is even, we have

$$\begin{aligned} \text{excess}(G_{t+t'}) &\geq \text{excess}(G_t) - 2t' \\ &> \text{excess}(G_t) - 2 \cdot \frac{\text{excess}(G_t)}{2} \\ &= 0 \end{aligned}$$

which is a contradiction, whereas in the case that $\text{excess}(G_t)$ is odd we have

$$\begin{aligned} \text{excess}(G_{t+t'}) &\geq \text{excess}(G_t) - 2(t' - 1) - 1 \\ &> \text{excess}(G_t) \\ &\quad - 2 \left(\frac{\text{excess}(G_t) + 1}{2} - 1 \right) - 1 \\ &= 0 \end{aligned}$$

which is also a contradiction, so we can conclude that Equation 3 holds. As in Theorem 2, the upper bound (Equation 4) is trivial. $\square$